```
# Transforms
base_transform = T.Compose(
    [
        T.Resize((64, 64)),
        T.ToTensor(),
        T.Normalize(mean=[0.485, 0.456, 0.406], std=[0.229, 0.224, 0.225]),
    ]
)

challenging_transform = T.Compose(
    [
        T.Resize((64, 64)),
        T.ColorJitter(brightness=0.5, contrast=0.5),
        T.RandomAffine(degrees=15, translate=(0.1, 0.1)),
        T.GaussianBlur(kernel_size=3),
        T.ToTensor(),
        T.Normalize(mean=[0.485, 0.456, 0.406], std=[0.229, 0.224, 0.225]),
    ]
)
```

Figure 12 | Environment noise simulation implementation.

```
def calculate_ers(model, normal_loader, challenging_loader):
    model.eval()
    with torch.no_grad():
        # Normal conditions
        correct_normal = 0
        total_normal = 0
        for images, labels in normal_loader:
            images, labels = images.to(device), labels.to(device)
            outputs = model(images)
            _, predicted = torch.max(outputs.data, 1)
            total_normal += labels.size(0)
            correct_normal += (predicted == labels).sum().item()
        normal_acc = correct_normal / total_normal

        # Challenging conditions
        correct_challenging = 0
        total_challenging = 0
        for images, labels in challenging_loader:
            images, labels = images.to(device), labels.to(device)
            outputs = model(images)
            _, predicted = torch.max(outputs.data, 1)
            total_challenging += labels.size(0)
            correct_challenging += (predicted == labels).sum().item()
        challenging_acc = correct_challenging / total_challenging

    ers = (challenging_acc / normal_acc) * 100
    return ers
```

Figure 13 | Environmental Robustness Score calculation.

C.3.4. Workshop Reviews

Reviewer #1: Review of Real-World Challenges in Pest Detection Using Deep Learning: An Investigation into Failures and Solutions.

```
Summary
This paper studies deep learning methods in pest detection applications. It highlights the need
for more research and attempts to perform first experiments in this area.

Results
The paper performs experiments on an image classifier, a ResNet-18 trained on ImageNet, fine-
tuned on the Crop Pest and Disease dataset. It uses various data augmentations to emulate the
real-world conditions and further trains the model using "multi-dataset training". The model
performance is measured in accuracy, loss, and Environmental Robustness Score (ERS). The first
experiment investigates the effect tuning the learning rate has on training. The paper claims
that a lower learning rate leads to a higher generalisation. The second experiment investigates
```